

Third CERP



Alternative Paradigms of Hardware Security for Addressing Threats of IP Piracy and Trojan during HLS

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Proposed Solutions of Research Problems

Sr. No.	Proposed Solutions of Research Problems	Journal Publication	Conference publication
1.	Robust security of hardware accelerators using protein molecular biometric signature and facial biometric encryption key	IEEE Transactions on Very Large Scale Integration Systems (TVLSI)	Proceedings of 8th IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST), China
2.	Securing Reusable IP Cores using Voice Biometric based Watermark	IEEE Transactions on Dependable and Secure Computing (TDSC)	--
3.	Exploring handwritten signature image features for hardware security	IEEE Transactions on Dependable and Secure Computing (TDSC)	--
4.	Secure hardware IP of GLRT cascade using color interval graph based embedded fingerprint for ECG detector	Nature Scientific Reports	Pending
5.	M-HLS: Malevolent High-Level Synthesis for Watermarked Hardware IPs	IEEE Embedded Systems Letters (ESL)	--
6.	Watermarking Hardware IPs Using Design Parameter Driven Encrypted Dispersion Matrix With Eigen Decomposition Based Security Framework	IEEE Access	Pending
7.	A Survey of High Level Synthesis based Hardware Security Approaches for Reusable IP Cores	IEEE Circuits and Systems Magazine (CASM)	--
8.	Palmprint Biometric vs Encrypted Hash based Digital Signature for Securing DSP Cores Used in CE systems	IEEE Consumer Electronics (CEM)	--
9.	PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis	Elsevier Journal on Expert Systems with Applications	Proceedings of 9th IEEE International Symposium on Smart Electronic Systems (IEEE – iSES), India
10.	Exploration of optimal functional Trojan-resistant hardware intellectual property (IP) core designs during high level synthesis	Elsevier Journal on Microprocessors and Microsystems	Pending
11.	Exploration of optimal crypto-chain signature embedded secure JPEG-CODEC hardware IP during high level synthesis	Elsevier Journal on Microprocessors and Microsystems	Proceedings of 35th IEEE International Conference on Microelectronics (ICM), Abu Dhabi
12.	Quadruple phase watermarking during high level synthesis for securing reusable hardware intellectual property cores	Elsevier Journal on Computers and Electrical Engineering	--
Total		12 Journals	3 Conferences

Hardware systems (Application specific systems)



- The rapid evolution of technology and the increasing complexity of computational tasks have underscored the growing need for specialized computing, also known as application-specific computing.
- This approach focuses on designing computing systems tailored to perform specific tasks more efficiently than general-purpose systems. Some of the factors leading to the rise in the need for specialized computing are (a) *Performance Optimization*, (b) *Energy Efficiency*, (c) *Cost Efficiency*, (d) *Industry-Specific Applications*, (e) *Customization and Flexibility*, etc.
- Some crucial examples of application-specific computing that includes data and computation-intensive operations are (a) *image processing applications/filters*, (b) *machine learning/deep learning-based applications*, (c) *JPEG compression-decompression*, (d) *DCT*, (E) *FIR filter*, etc.
- These computation-intensive applications are designed as dedicated reusable hardware *Intellectual Property (IP) core* of *Hardware Accelerator* using the High Level Synthesis (HLS) process [1].

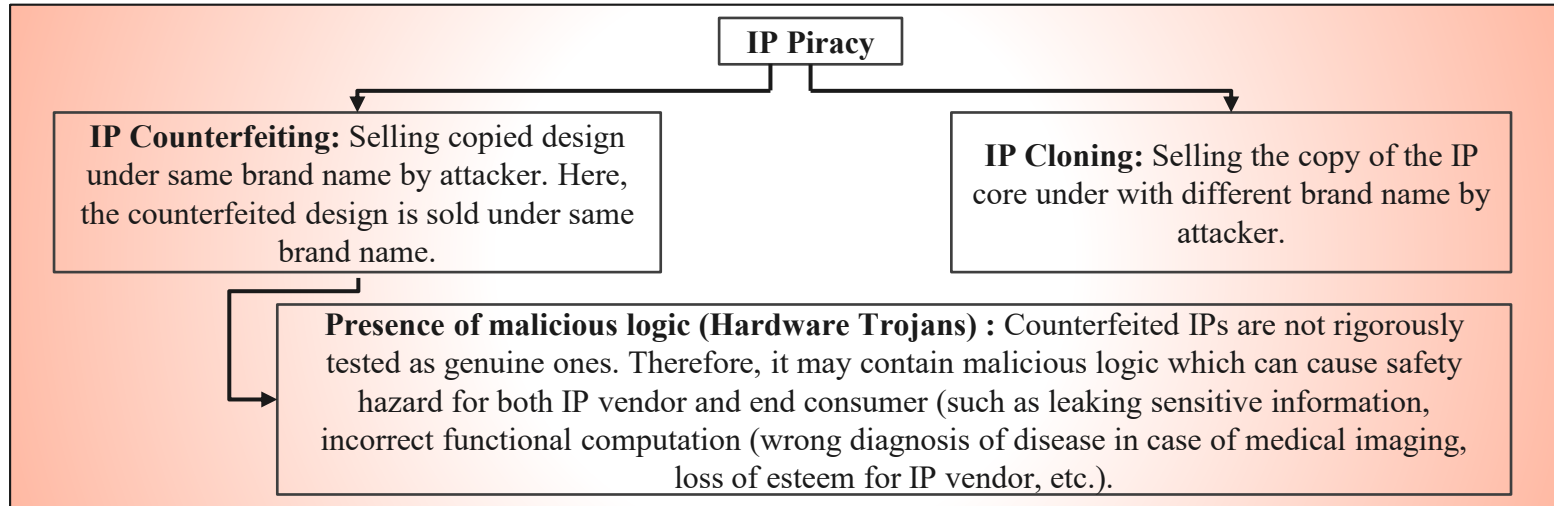
Hardware design process

- Due to the globalization of the design supply chain, the design process of these application-specific hardware systems involves various hardware threats [1], [2].



Figure 1: Hardware IC design process

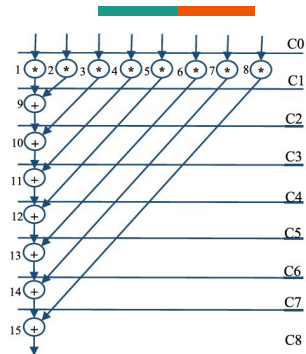
Security Issues associated with hardware IP Cores [1], [2]



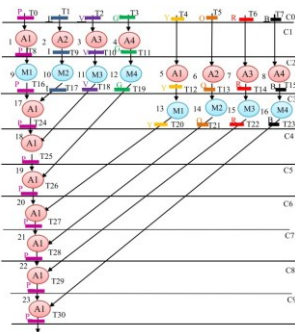
Fraudulent claim of IP ownership: An adversary tries to fraudulently claim the ownership of the IP.

Therefore, it is essential to secure these image processing filter IP cores from these hardware threats.

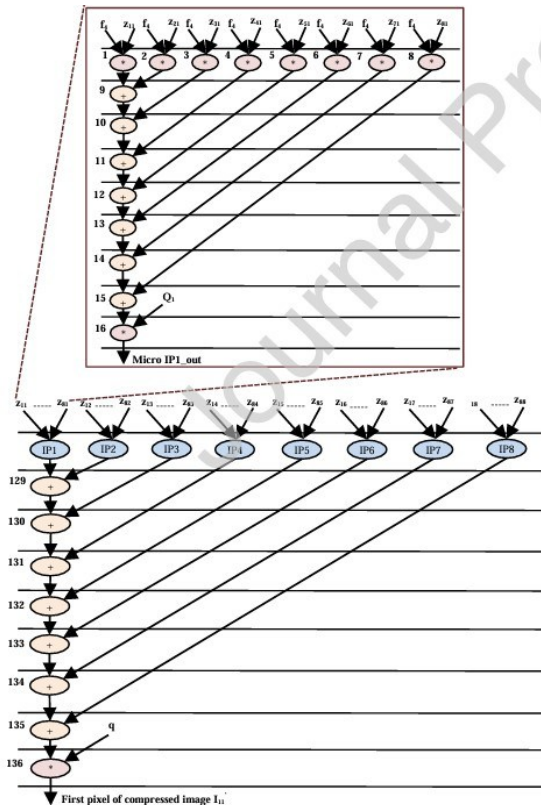
Benchmarks



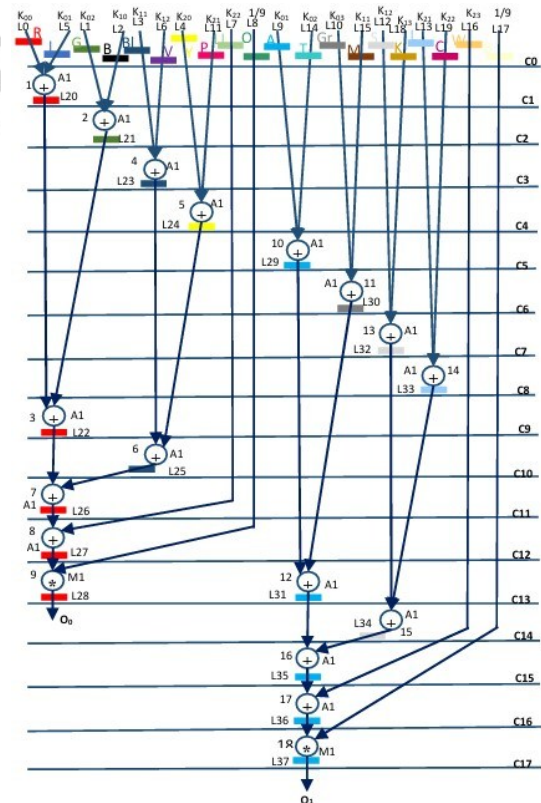
Data flow graph of 8-point DCT



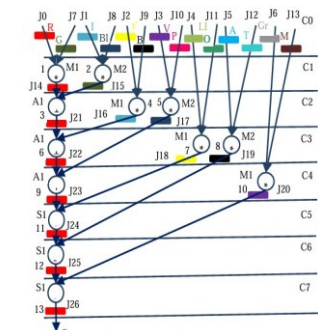
Data flow graph of FIR filter



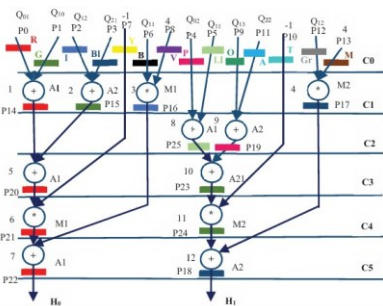
Data flow graph of JPEG-CODEC



Data flow graph of blur filter



Data flow graph of IIR



Data flow graph of Laplacian filter

Related Work

Sr. No.	Existing Work	Technique Used	Remarks
1.	a). F. Koushanfar, I. Hong, and M. Potkonjak [3] (2005) b). A. Sengupta and S. Bhadauria [4] (2016)	IP seller's signature encoding-based hardware watermarking.	Weak watermarking mechanism due to involvement of only two and four variable signature encoding process. Not robust and future proof. The watermark inserted by watermarking technique becomes vulnerable if relevant information (like signature size, digit encoding, and combination) gets leaked.
2.	a) A. Sengupta and M. Rathor, [5] (2019) b). E. Castillo, et. al., [6] (2008) c). A. Sengupta, et. al., [7] (2019)	Steganography, encryption and digital signature-based hardware watermarking.	Provides weaker security due to the generation of limited security constraints. Additionally, they become weak in case of a compromised stego-threshold and RSA key value.
3.	a). A. Sengupta and M. Rathor, [8] (2020) b). A. Sengupta, et. al., [9] (2021)	IP seller's biometrics (such as facial and palmprint biometrics) based hardware watermarking.	Although [8] and [9] integrate the natural identity of IP sellers with the hardware design, they provide weaker security due to the generation of limited watermarking (security) constraints.

Secure hardware IP of GLRT cascade using color interval graph based embedded fingerprint for ECG detector

- The proposed approach presents a novel secure hardware IP of generalized likelihood ratio test (GLRT) cascade using color interval graph (CIG) based embedded fingerprint, for electrocardiogram (ECG) detector, for the first time in literature [10].
- The GLRT unit is the primary component of the ECG detector that executes computation-intensive functions to estimate heart rate using (*Q wave*, *R wave* and *S wave*) *QRS wave complex* from incoming filter signals.
- The proposed approach uses the IP seller's fingerprint biometric sample to generate and embed the watermark signature (watermarking constraints) into the final hardware IP design.
- The embedded IP seller's watermark helps in the demarcation and isolation of pirated IPs from the authentic ones by system-on-chip (SoC) integrator before integration into the final hardware system.

Motivation

- From the perspective of the end user (patient), the safe and reliable functioning of the GLRT unit in the ECG detector is critical as it is responsible for the generation of important ECG parametric data such as *Heart Rate (HR)*, *PR Interval (PRI)*, *QRS Interval (QRSI)*, *QT Interval (QTI)*, *QTC Interval (QTCI)*.
- Therefore, it is essential to design the GLRT unit of the ECG detector as a reusable hardware IP core because of its wide usability and computation-intensive nature.
- A pirated (*i.e.*, counterfeited) GLRT hardware IP core is unreliable and may contain malicious logic that could result into inaccurate detection of vital ECG parametric data, erratic behavior or functionality of the ECG detector, mistimed pulse from ECG detector for cardiac pacemaker devices. Therefore, it is necessary to secure the hardware IP so that it can be verified before integration into the final system.

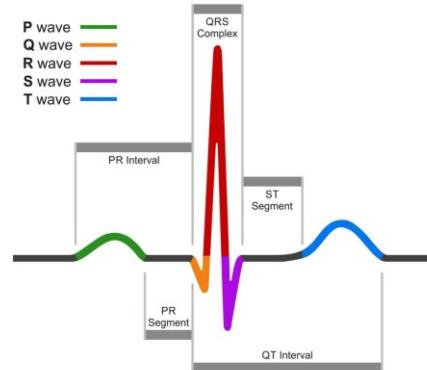


Figure 2: Sample ECG waveform

Detailed of the proposed approach

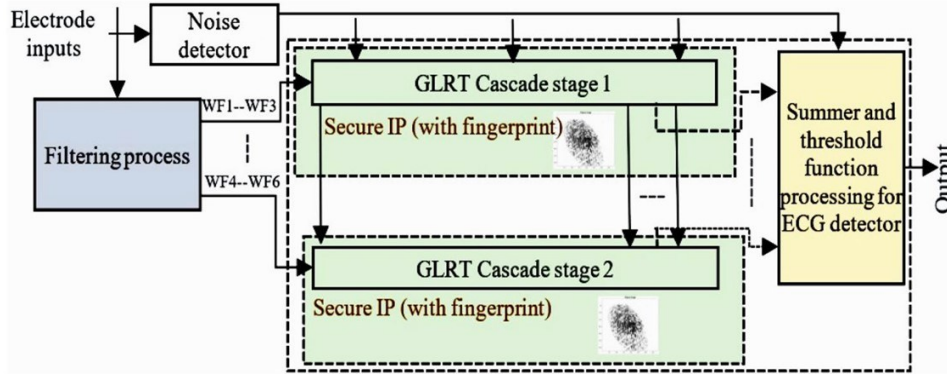


Figure 3: Proposed secure hardware IP of GLRT cascade for ECG detector

- At first, GLRT's mathematical/transfer function is taken as input, which is converted into its respective control data flow graph (CDFG).
- Next, the obtained CDFG is scheduled using input resource constraints (*i.e.*, number of adders and multipliers) to generate its corresponding scheduled data flow graph (SDFG).
- Transfer function: $Z(a) = s^T(n)H(H^TH)^{-1}H^Ts(n)$, where $s(n)$ is the input to the filtering unit and H is the linear combination matrix of the representative function. Here, $s^T(n)$ is a 6-by-1 matrix, H is a 1-by-6 matrix, $s(n)$ is a 1-by-6 matrix and (H^TH) is a 6-by-6 matrix.

Details of fingerprint watermark signature generation process

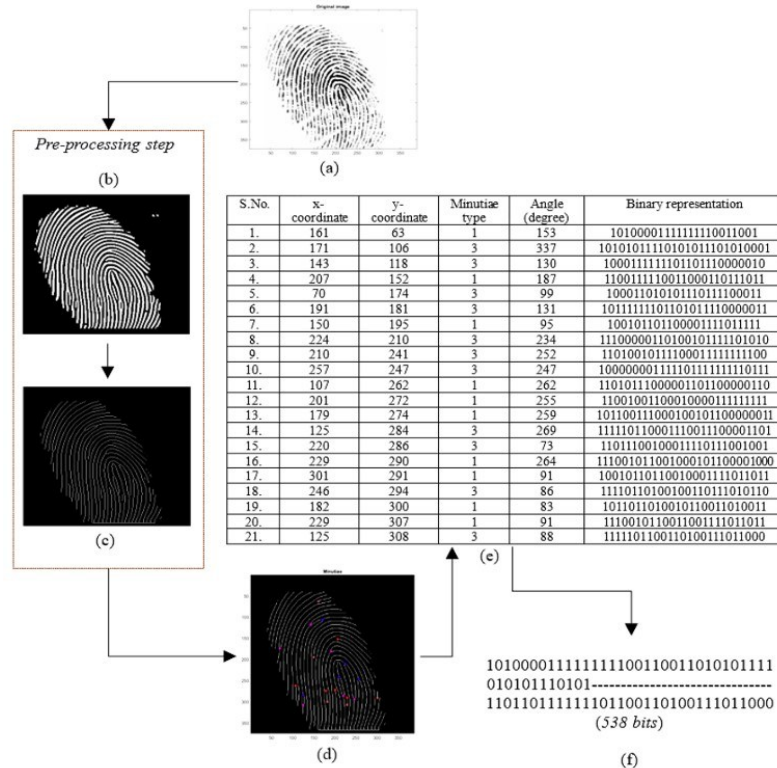


Figure 4: Proposed fingerprint digital template generation process extracted from captured IP vendor's fingerprint, (a) input IP seller's fingerprint image, (b) binarized fingerprint image, (c) thinned fingerprint image, (d) minutiae points generation on fingerprint image, (e) details of generated minutiae points parameters, (f) generated fingerprint biometric based digital template. The biometric captured is of a real IP vendor entity that is used for further processing of template generation

Final watermark constraints embedded SDFG of GRLT

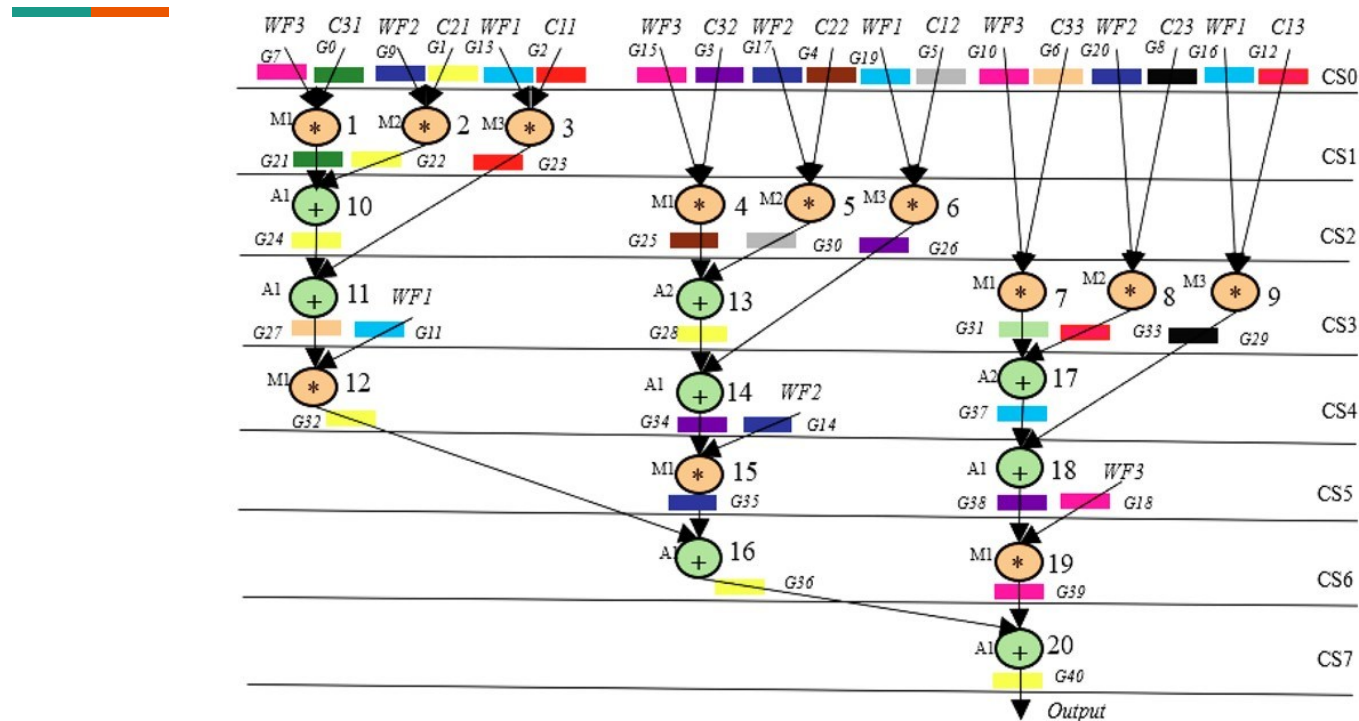


Figure 5: SDFG of GLRT cascade macro IP scheduled using three multipliers and two adders post embedding fingerprint

Final watermark constraints embedded RTL

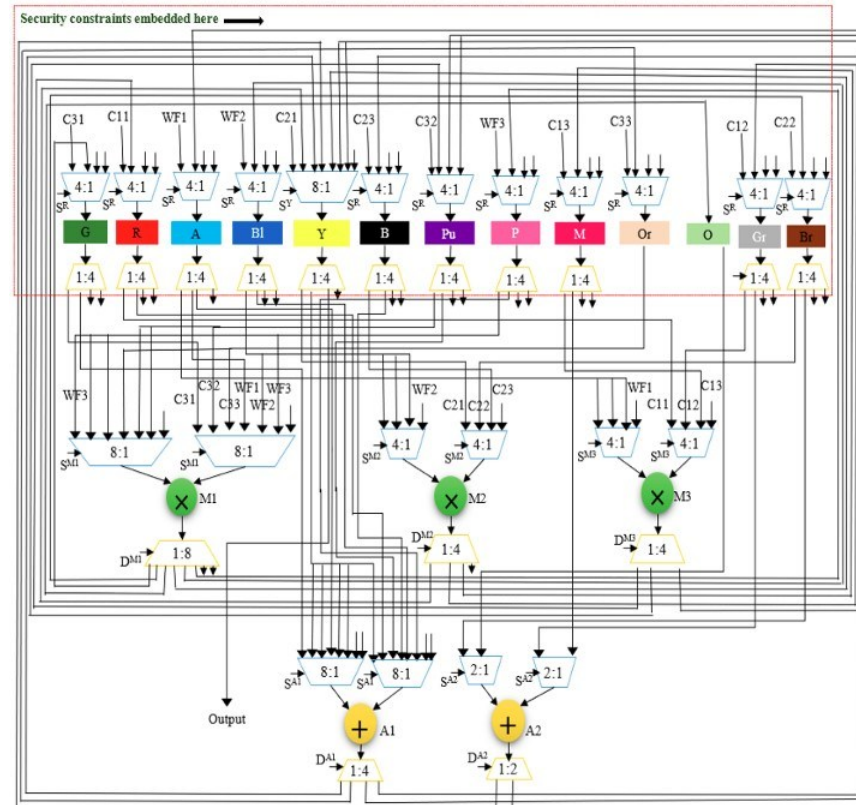


Figure 6: Secure RTL design of GLRT cascade macro IP core with CIG based embedded fingerprint

Evaluation parameters

- **Evaluation of Robustness Using Probability of Coincidence (P_c/C_i):**

$$P_c = \left(1 - \frac{1}{c}\right)^f$$

‘ c ’ denotes the number of registers used in the RAT, and ‘ f ’ denotes the number of hardware constraints added.

- **Evaluation of tamper tolerance (TT/T_i):**

$$TT = (w)^f$$

‘ w ’ is the number of types of digits in the signature, and ‘ f ’ is the signature size (or the number of corresponding hardware security constraints)

- **Design cost:**

$$Design\ cost = q1 * \left(\frac{Area(A)}{A_{max}}\right) + q2 * \left(\frac{Latency\ (L)}{L_{max}}\right)$$

where $q1=0.5$ and $q2=0.5$ are designer-defined weighing factors used to provide equal weightage to design *area* (A) and execution time (*Latency* (L)) during design cost function evaluation. Further, A_{max} and L_{max} represents maximum design area (determined with available maximum functional resources) and time (delay) (determined with available minimum functional resources)

Results

Table 1: Comparison of tamper tolerance (TT) between the proposed fingerprint embedded secure GLRT cascade IP with facial biometric [8] embedded IP design and digital signature embedded IP design [7]

Proposed secure GLRT IP with fingerprint		Design with facial constraints [25]		Design with digital signature [16]	
Security constraints	T_i	Security constraints	T_i	Security constraints	T_i
250	1.80E+75	16	6.55E+04	16	6.55E+04
275	6.07E+82	32	4.29E+09	32	4.29E+09
300	2.03E+90	64	1.84E+19	64	1.84E+19
346	1.43E+104	81	2.41E+24	128	3.40E+38

Table 3: Comparison of probability of coincidence between the proposed fingerprint embedded secure GLRT cascade IP with facial biometric [8] embedded IP design and digital signature embedded IP design [7]

Proposed secure GLRT IP with fingerprint		Design with facial constraints [25]		Design with digital signature [16]	
Security constraints	C_i	Security constraints	C_i	Security constraints	C_i
250	3.57E-10	16	2.48E-01	16	2.48E-01
275	4.05E-11	32	6.17E-02	32	7.71E-02
300	4.60E-12	64	3.81E-03	64	3.81E-03
346	8.41E-14	81	8.69E-04	128	1.45E-05

Table 2: Comparison of tamper tolerance (TT) between the proposed fingerprint embedded secure GLRT cascade IP with encrypted signature embedded IP design [6] and hardware watermarking embedded IP design [3]

Proposed secure GLRT IP with fingerprint		Design with encrypted signature [17]		Design with watermark [12]	
Security constraints	T_i	Security constraints	T_i	Security constraints	T_i
250	1.80E+75	32	6.17E-02	32	6.17E-02
275	6.07E+82	64	3.81E-03	64	3.81E-03
300	2.03E+90	128	1.45E-05	128	1.45E-05
346	1.43E+104	160	8.99E-07	240	8.52E-10

Table 4: Comparison of probability of coincidence between the proposed fingerprint embedded secure GLRT cascade IP with facial biometric [8] embedded IP design and digital signature embedded IP design [7]

Proposed secure GLRT IP with fingerprint		Design with encrypted signature [17]		Design with watermark [12]	
Security constraints	C_i	Security constraints	C_i	Security constraints	C_i
250	3.57E-10	32	6.17E-02	32	6.17E-02
275	4.05E-11	64	3.81E-03	64	3.81E-03
300	4.60E-12	128	1.45E-05	128	1.45E-05
346	8.41E-14	160	8.99E-07	240	8.52E-10

Results (Design cost)

Table 5: Design latency, area, and resource configuration of proposed secure GLRT IP before and after embedding fingerprint signature

Application	Resource configuration	Unsecured design (before fingerprint embedding)		Proposed fingerprint embedded secure design	
GLRT cascade hardware IP core	2(+), 3(*)	Design area (um ²)	Design latency (ps)	Design area (um ²)	Design latency (ps)
		273.67	1656.07	273.67	1656.07

Table 6: Design cost, leakage power, register count and resource configuration of proposed secure GLRT hardware IP before and after embedding fingerprint signature

Application	Resource configuration	Unsecured design (before fingerprint embedding)		Proposed fingerprint embedded secure design	
GLRT cascade hardware IP core	2(+), 3(*) and 13 registers	Design cost	Leakage power	Design cost	Leakage power
		0.43	8.57 μw	0.43	8.57 μw

Robust security of hardware accelerators using protein molecular biometric signature and facial biometric encryption key

- This proposed approach presents a novel molecular biometric-based hardware security approach based on IP seller's protein molecule sequence, for the first time to secure hardware IP cores [11].
- Here, an IP seller/vendor-selected protein sequence comprising 20 unique amino acid combinations is used for molecular signature generation.
- The generated signature is then encrypted through AES using an encryption key generated with the facial biometric of an authentic IP vendor. Thus, the proposed approach incorporates two classes of biometrics of IP seller to ensure highly robust and unique authentication.
- The encrypted molecular signature is then converted into watermarking constraints using the IP seller's mapping rule and is further embedded in the hardware design during the register allocation phase of the HLS process.
- The embedded IP seller's watermark helps in the demarcation and isolation of pirated IPs from the authentic ones and protects IP seller from false IP ownership assertions.

Detailed of the proposed approach

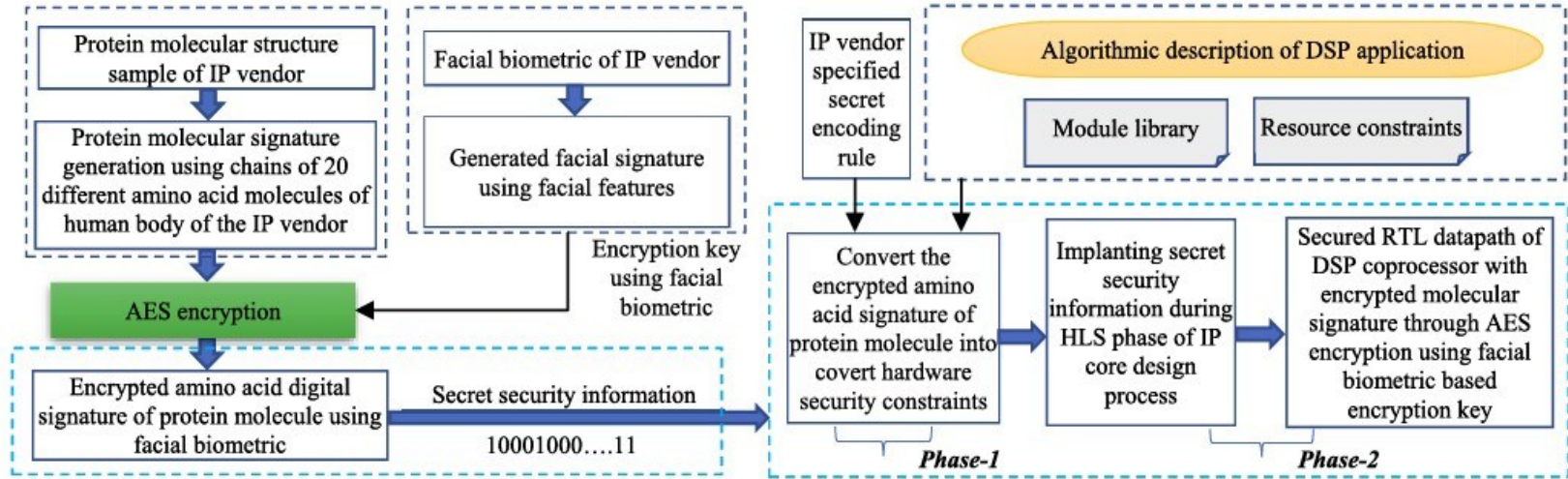


Figure 7: Overview of the proposed methodology

Details of protein molecular signature generation process

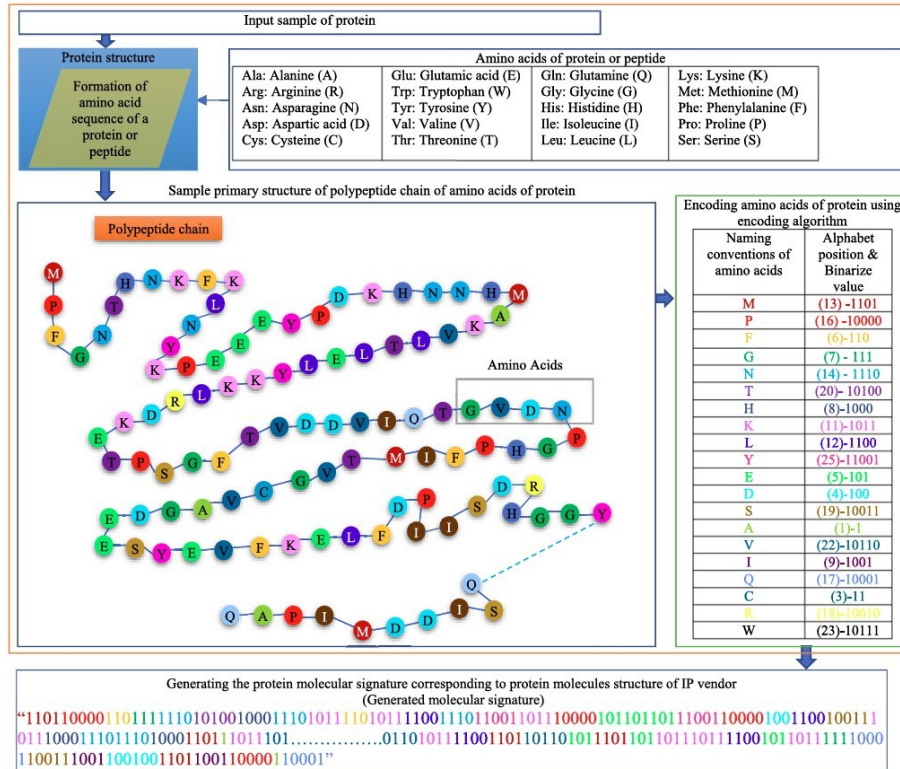


Figure 8: Generating the protein molecular signature corresponding to the amino acid sequence of sample protein

Details of IP seller's facial biometric template generation process

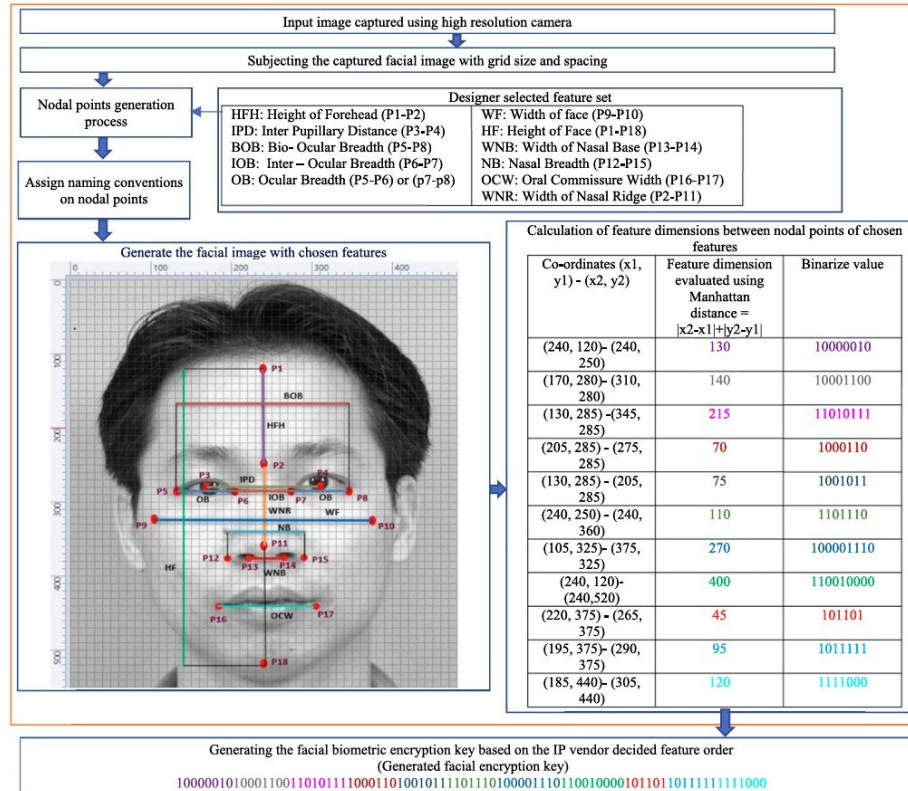


Figure 9: Demonstration of facial biometric key generation used for encrypting the protein molecular signature

Details of IP seller's facial biometric template generation process

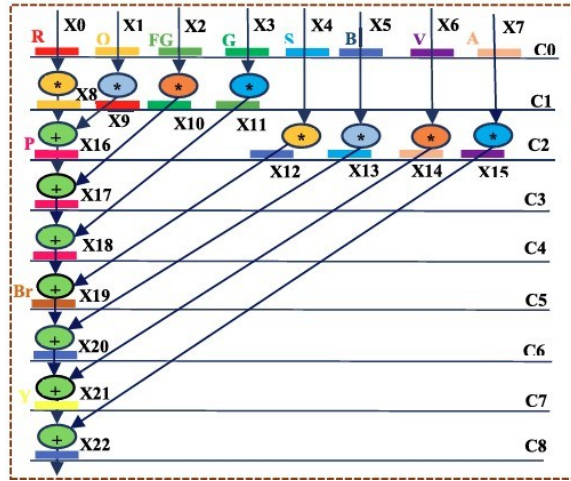


Figure 10: Scheduled DFG of DCT-8 with 1(+) and 4(*) post embedding secret constraints

Table 7: Register allocation table (RAT) corresponding to 8-point DCT application after embedding watermarking constraints

CS	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11
CS0	X0	X1	X2	X3	X4	X5	X6	X7	--	--	--
CS1	X9	X8	X11	X10	X4	X5	X6	X7	--	--	--
CS2	--	--	X11	X10	X13	X12	X15	X14	X16	--	--
CS3	--	--	X11	--	X13	X12	X15	X14	X17	--	--
CS4	--	--	--	--	X13	X12	X15	X14	X18	--	--
CS5	--	--	--	--	X13	--	X15	X14	--	X19	--
CS6	--	--	--	--	--	X20	X15	X14	--	--	--
CS7	--	--	--	--	--	--	X15	--	--	--	X21
CS8	--	--	--	--	--	X22	--	--	--	--	--

Results

Table 8: comparison of security in terms of Pc for JPEG-CODEC IP core between proposed approach and IP fingerprinting [14]

Fingerprint image	# of embedded security constraints of fingerprint approach	Pc of fingerprint approach [14]	Proposed amino acid chain of protein sequence	# of embedded security constraints of proposed approach	Pc of proposed approach	% Reduction of Pc obtained using proposed approach
Original Image:101_1	350	8.0E-3	150	599	2.5E-4	96.8%
Original Image:101_2	418	3.1E-3	200	799	1.6E-5	99.4%
Original Image:101_8	526	7.0E-4	250	990	1.1E-6	99.8%
Original Image:102_3	538	5.9E-4	300	1184	8.0E-8	99.9%
Original Image:103_8	555	4.7E-4	350	1382	5.2E-9	99.9989%

Table 9: Comparison of security in terms of TT for JPEG-CODEC IP core between proposed approach and IP fingerprinting [14]

Fingerprint image	# of embedded security constraints of fingerprint approach	TT of fingerprint approach [14]	Proposed amino acid chain of protein sequence	# of embedded security constraints of proposed approach	TT of proposed approach	% Increment of TT obtained using proposed approach
Original Image:101_1	350	2.29E+105	150	599	2.07E+180	9.0393E+76%
Original Image:101_2	418	6.76E+125	200	799	3.33E+240	4.92604E+116%
Original Image:101_8	526	2.19E+158	250	990	1.04E+298	4.74886E+141%
Original Image:102_3	538	8.99E+161	300	1184	~1.0E+358	~+198%
Original Image:103_8	555	1.17E+167	350	1382	~1.0E+417	~+253%

Results

Table 10: Comparison of Pc w.r.t. related work [15]

Bench- marks	Proposed		Related work [15]	
	Max. constraints	Pc	Max. constraints	Pc
FIR	225	0.9E-13	128	3.7E-8
ARF	306	1.79E-18	128	3.7E-8
DWT	110	2.1E-11	92	1.2E-9
JPEG	1408	3.6E-9	128	1.7E-1
MESA	1408	1.3E-13	128	3.7E-8

Table 11: Comparison of tamper tolerance (TT) w.r.t. related work [15]

Bench- marks	Proposed		Related work [15]	
	Max. constraints	TT	Max. constraints	TT
FIR	225	5.39E+67	128	3.40E+38
ARF	306	1.30E+92	128	3.40E+38
DWT	110	1.29E+33	92	4.95E+27
JPEG	1408	1.0E+421	128	3.40E+38
MESA	1408	1.0E+421	128	3.40E+38

Table 12: Design cost of embedding encrypted protein molecular signature

Benchmarks	Design cost of encrypted protein molecular signature implanted design corresponding to Sequence-1 (1408 digits)	Design cost of encrypted protein molecular signature implanted design corresponding to Sequence-2 (128 digits)
8-point DCT	0.473	0.473
FIR	0.569	0.567
ARF	0.476	0.473
DWT	0.615	0.617
JPEG	0.214	0.214
MESA	0.280	0.280

Watermarking Hardware IPs Using Design Parameter Driven Encrypted Dispersion Matrix With Eigen Decomposition Based Security Framework



- This paper presents a mathematical watermarking methodology using a design parameter-driven encrypted dispersion matrix (*characteristics of the IP vendor selected design space parameters*) with an eigen decomposition-based security framework (*design space's characteristics in terms of IP vendor chosen resource configuration*) for protecting hardware IP cores [12].
- An encrypted mathematical watermark is generated using dispersion matrix, eigen decomposition, and AES encryption.
- The encrypted watermark is then converted into watermarking constraints using the IP vendor's mapping rule and is further embedded in the hardware design during the register allocation phase of the HLS process.
- The embedded IP seller's watermark provides a detective countermeasure against potential IP piracy and false IP ownership claim by an adversary in SoC and fabrication house.

Detailed of the proposed approach

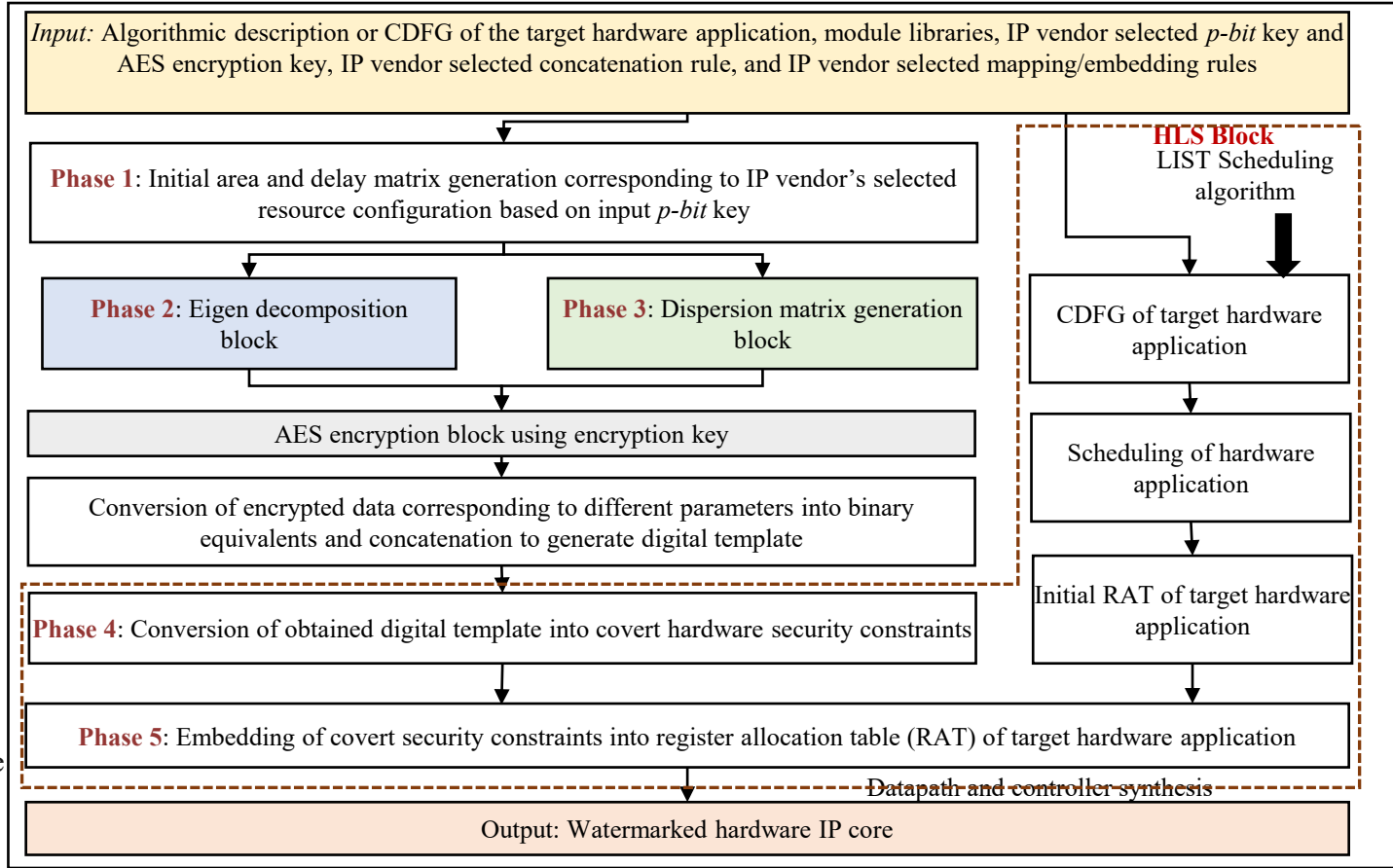


Figure 11:
Overview of the
proposed
methodology

Generation of initial data

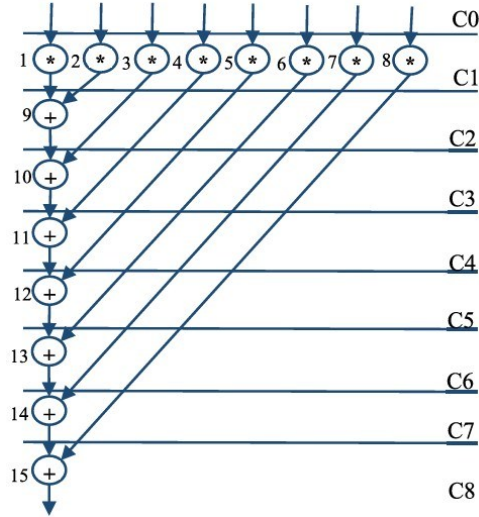


Figure 12: Control data flow graph of 8-point DCT

p -bit key	R_C (IP vendor chosen - key controlled)	A_d (IP vendor computed)	L_d (IP vendor computed)
011	[1, 4]	327 μm^2	927 ps
000	[1, 1]	101 μm^2	2186 ps
111	[1, 8]	629 μm^2	729 ps
100	[1, 5]	403 μm^2	927 ps

Step 1. Computation of area (A_d) and latency (L_d) corresponding to IP vendor chosen resource configuration:

The area and latency corresponding to selected resource configurations are shown in Table 1.

Step 2. (a). Mean computation of design parameter ' A_d ':

$$\overline{A_d} = \sum_{i=1}^n A_{di}$$

$$\overline{A_d} = \frac{(327 + 101 + 629 + 403)}{4} = 365 \quad (1)$$

Step 2. (b). Mean computation of design parameter ' L_d ':

$$\overline{L_d} = \sum_{i=1}^n L_{di}$$

$$\overline{L_d} = \frac{(927 + 2186 + 729 + 927)}{4} = 1192.25 \approx 1192 \quad (2)$$

Estimation of dispersion matrix

Step 3. (a). Subtract the mean ($\bar{A}_d$) from all area parameter values:

⇒

$$(A_{d1} - \bar{A}_d), (A_{d2} - \bar{A}_d), \\ (A_{d3} - \bar{A}_d), \dots, (A_{dn} - \bar{A}_d) \quad (3)$$

$$\Rightarrow (327-365), (101-365), (629-365), (403-365)$$

$$\Rightarrow (-38), (-264), (264), (38)$$

Step 3. (b). Subtract the mean ($\bar{L}_d$) from all latency parameter values:

⇒

$$(L_{d1} - \bar{L}_d), (L_{d2} - \bar{L}_d), \\ (L_{d3} - \bar{L}_d), \dots, (L_{dn} - \bar{L}_d) \quad (4)$$

$$\Rightarrow (927-1192), (2186-1192), (729-1192), (927-1192)$$

$$\Rightarrow (-265), (994), (-463), (-265)$$

Step 4. (a). Compute the sum of the square of the differences corresponding to the design area:

$$S_A = \sum_{i=1}^n (A_{di} - \bar{A}_d)^2 \quad (5)$$

$$\Rightarrow S_A = (-38)^2 + (-264)^2 + (264)^2 + (38)^2$$

$$\Rightarrow (1444 + 69696 + 69696 + 1444) = 142280$$

Step 4. (b). Compute the sum of the square of the differences corresponding to design latency:

$$S_L = \sum_{i=1}^n (L_{di} - \bar{L}_d)^2 \quad (6)$$

$$\Rightarrow S_L = (-265)^2 + (994)^2 + (-463)^2 + (-265)^2$$

$$\Rightarrow (70225 + 988036 + 214369 + 70225) = 1342855$$

Step 5. Estimate $\text{var}(A_d)$, $\text{var}(L_d)$, and $\text{cov}(A_d, L_d)$:

$$\text{Var}(A_d) = \frac{\sum_{i=1}^n (A_{di} - \bar{A}_d)^2}{n-1} \quad (7)$$

$$\text{Var}(A_d) = \left(\frac{142280}{3} \right) = 47426.66 \approx 48000$$

$$\text{Var}(L_d) = \frac{\sum_{i=1}^n (L_{di} - \bar{L}_d)^2}{n-1} \quad (8)$$

$$\text{Var}(L_d) = \left(\frac{1342855}{3} \right) = 447618.33 \approx 448000$$

$$\text{Cov}(A_d, L_d) = \sum_{i=1}^n \frac{(A_{di} - \bar{A}_d) \times (L_{di} - \bar{L}_d)}{n-1} \quad (9)$$

Now, perform the multiplication of the corresponding pair's values obtained in steps 3. (a) and 3. (b).

$$\Rightarrow \{(-38) \times (-265)\}, \{(-264) \times (994)\}, \{(264) \times (-463)\}, \\ \{(38) \times (-265)\}$$

$$\Rightarrow \{10070\}, \{-262416\}, \{-122232\}, \{-10070\}$$

Next, perform a summation of the above-obtained values to estimate $\text{Cov}(A_d, L_d)$.

$$\Rightarrow (10070 - 262416 - 122232 - 10070)$$

$$\Rightarrow (-384648)$$

$$\Rightarrow \text{Cov}(A_d, L_d) = \left(\frac{-384648}{4-1} \right) = -128216$$

(7) Finally, the generated dispersion matrix is:

Dispersion matrix (DM)

$$= \begin{bmatrix} \frac{\sum_{i=1}^n (A_{di} - \bar{A}_d)^2}{n-1} & \sum_{i=1}^n \frac{(A_{di} - \bar{A}_d) \times (L_{di} - \bar{L}_d)}{n-1} \\ \sum_{i=1}^n \frac{(A_{di} - \bar{A}_d) \times (L_{di} - \bar{L}_d)}{n-1} & \frac{\sum_{i=1}^n (L_{di} - \bar{L}_d)^2}{n-1} \end{bmatrix}$$

$$DM = \begin{bmatrix} \text{Var}(A_d) & \text{Cov}(A_d, L_d) \\ \text{Cov}(A_d, L_d) & \text{Var}(L_d) \end{bmatrix}$$

$$DM = \begin{bmatrix} 48000 & -128216 \\ -128216 & 448000 \end{bmatrix}$$

Estimation of eigen roots

$$A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 8 \\ 1 & 5 \end{bmatrix}$$

$$\Rightarrow \det(\lambda I - A) = 0$$

$$\Rightarrow \det \left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} \lambda - 1 & -4 \\ -1 & \lambda - 1 \end{bmatrix} \right) = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0$$

$$\Rightarrow \lambda_1 = 3 \text{ and } \lambda_2 = -1$$

Similarly,

$$\Rightarrow \det(\lambda I - B) = 0$$

$$\Rightarrow \det \left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 8 \\ 1 & 5 \end{bmatrix} \right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} \lambda - 1 & -8 \\ -1 & \lambda - 5 \end{bmatrix} \right) = 0$$

$$\Rightarrow \lambda^2 - 6\lambda - 3 = 0$$

$$\lambda_3 = 6.46 \text{ and } \lambda_4 = -0.46$$

Table 13: Register allocation table (RAT) pre and post embedding generated signature

CS	Red(R)	Green (G)	Indigo (I)	Blue (BL)	Yellow (Y)	Black (B)	Violet (V)	Pink (P)	Lime (LI)	Olive (O)	Aqua (A)	Teal (T)	Grey (Gr)	Magenta (M)	Silver (S)	Khaki (K)
0	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15
1	L16/L17	L17 /L16	L2	L3	L4	L5	L6	L7	-	-	-	-	-	-	-	-
2	L24	-	L18 /L19	L19 /L18	L4	L5	L6	L7	-	L24	-	-	-	-	-	-
3	L25	-	L19	L19	L20 /L21	L21 /L20	L6	L7	L25	-	-	-	-	-	-	-
4	L26	-	-	-	L20 /L21	L21 /L20	L22	L23	-	L26	-	-	-	-	-	-
5	L27	-	-	-	L21	L21	L22 /L23	L23 /L22	-	-	L27	-	-	-	-	-
6	L28	-	-	-	-	-	L22 /L23	L23 /L22	-	L28	-	-	-	-	-	-
7	L29	-	-	-	-	-	L23	L23	-	-	-	L29	-	-	-	-
8	L30	-	-	-	-	L30	-	-	-	-	-	-	-	-	-	-

Results

Table 14: Comparison of the probability of coincidence (C_i) between the proposed approach, [8], [6]

Benchmarks	Register count before embedding security constraints	Proposed approach		[8]		[6]	
		Embedded security constraints (c)	(C_i)	Embedded security constraints (c)	(C_i)	Embedded security constraints (c)	(C_i)
8-point DCT	16	214	1.00E-06	81	5.36E-03	160	3.27E-05
FIR	16	343	2.43E-10	81	5.36E-03	160	3.27E-05
ARF	16	441	4.35E-13	81	5.36E-03	160	3.27E-05
DWT	10	164	3.13E-08	81	1.96E-04	160	4.77E-08
JPEG-CODEC	137	896	1.41E-03	81	5.52E-01	160	3.09E-01

Table 15: Comparison of the probability of coincidence (C_i) between the proposed approach, [3], [14]

Benchmarks	Register count before embedding security constraints	Proposed approach		[3]		[14]	
		Embedded security constraints	(C_i)	Embedded security constraints	(C_i)	Embedded security constraints	(C_i)
8-point DCT	16	214	1.00E-06	128	2.58E-04	199	2.64E-06
FIR	16	343	2.43E-10	128	2.58E-04	199	2.64E-06
ARF	16	441	4.35E-13	128	2.58E-04	199	2.64E-06
DWT	10	164	3.13E-08	128	1.39E-06	164	3.13E-08
JPEG-CODEC	137	896	1.41E-03	128	3.91E-01	199	2.32E-01

Results

Table 16: Comparison of tamper tolerance (TT) between the proposed approach, [8], [6], [3], and

Benchmarks	Proposed approach	[8]	[6]	[3]	[14]
8-point DCT	2.63E+64	2.41E+24	1.46E+48	3.40E+38	8.03E+59
FIR	1.79E+103	2.41E+24	1.46E+48	3.40E+38	8.03E+59
ARF	5.67E+132	2.41E+24	1.46E+48	3.40E+38	8.03E+59
DWT	2.33E+49	2.41E+24	1.46E+48	3.40E+38	8.03E+59
JPEG-CODEC	5.28E+269	2.41E+24	1.46E+48	3.40E+38	8.03E+59

Table 17: Design cost of proposed approach pre and post implanting generated signature

Benchmarks	IP vendor selected resource configuration for scheduling	Initial design (<i>i.e.</i> , pre signature implanted unsecured design)				Final secured signature implanted design				Design cost overhead %
		Area (um ²)	Latency (ps)	Register count	Design cost	Area (um ²)	Latency (ps)	Design cost	Register count	
8-point DCT	1(+), 2(*)	182.45	1324.86	16	0.446	182.45	1324.86	0.446	16	0
FIR	1(+), 2(*)	106.95	2583.46	16	0.569	109.31	2583.46	0.57	19	0.17
ARF	1(+), 2(*)	182.45	2450.98	16	0.412	187.95	2450.98	0.415	23	0.72
DWT	2(+), 3(*)	272.10	1722.31	10	0.656	275.25	1722.31	0.657	14	0.15
JPEG-CODEC	6(+), 8(*)	824.96	3245.89	137	0.157	824.96	3245.89	0.157	137	0

PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis



- The proposed approach is based on multi-phase encryption technique used for securing image processing filters IP cores. This paper discusses signature-based hardware security methodology on image processing IP cores for the first time [13].
- The proposed approach uses the register allocation table of the image processing application to generate secret data, which is used for multi-phase encryption to determine hardware security constraints.
- The generated hardware security constraints then embedded in the image processing filter IP cores design to authenticate genuine IP seller.
- The huge variation in the key selection in different phases of encryption and the use of secret steganographic data for signature generation increases the robustness of the proposed hardware security methodology for image filters.

Detailed flow diagram of the proposed approach

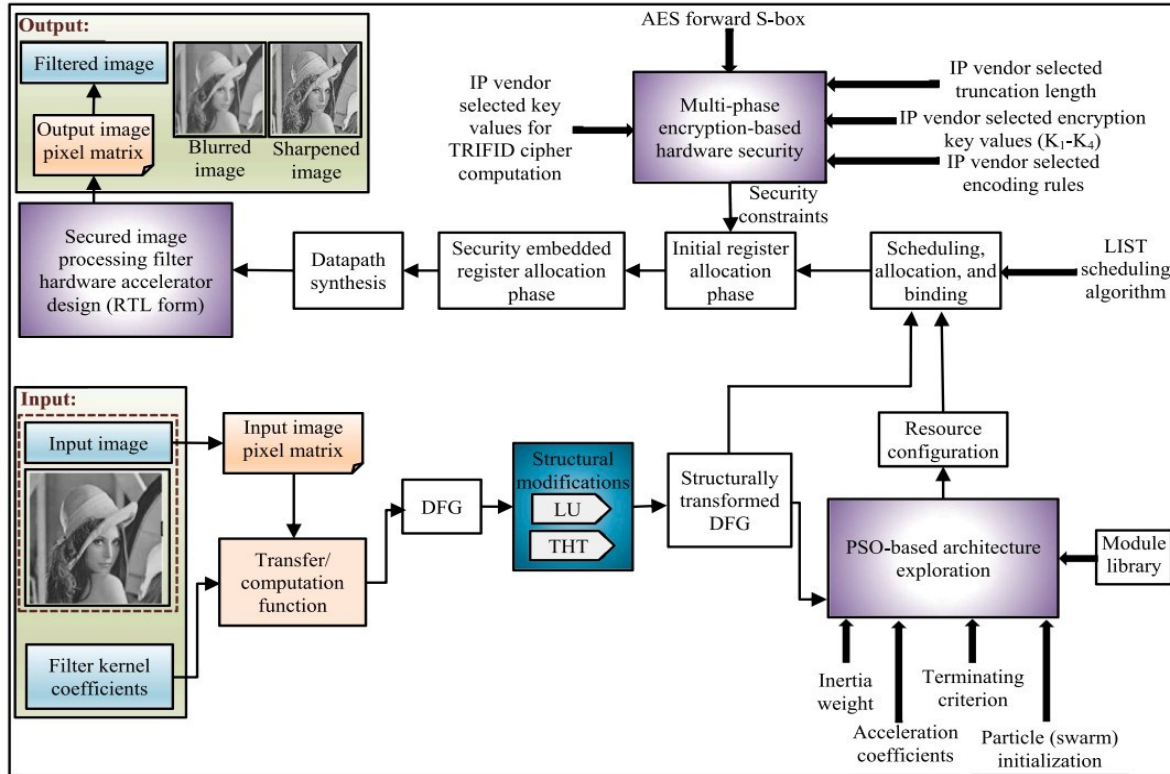


Figure 13: Figure illustrating the overview of the proposed low-cost multi-phase encryption-based hardware security for securing image processing filter IP cores

Details of multi-phase encryption based hardware security methodology

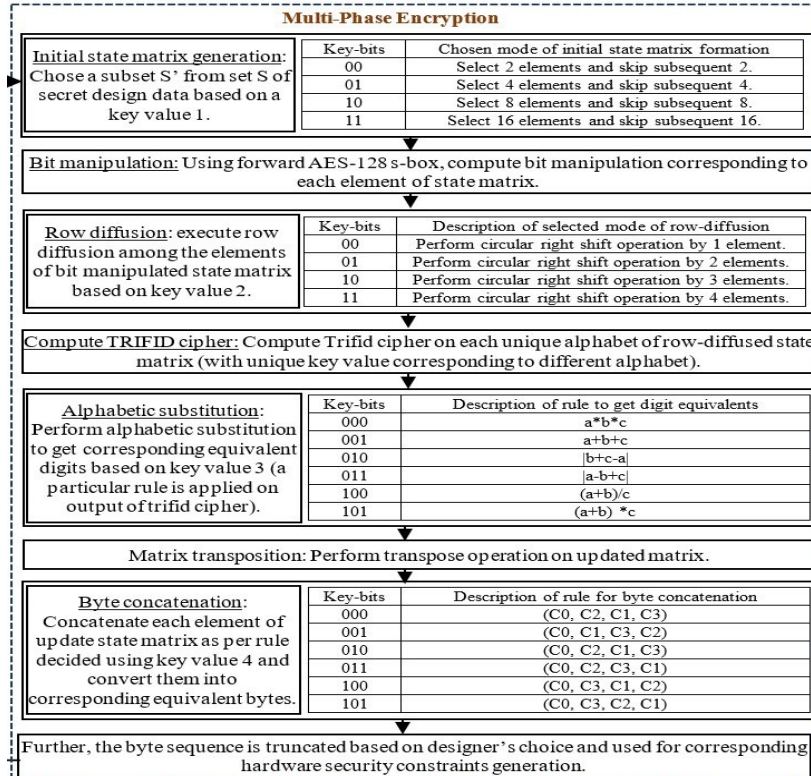


Figure 14: Flow-chart of proposed methodology for securing image processing filters using multi-phase encryption

Extraction of secret design data from SDFG of image processing application

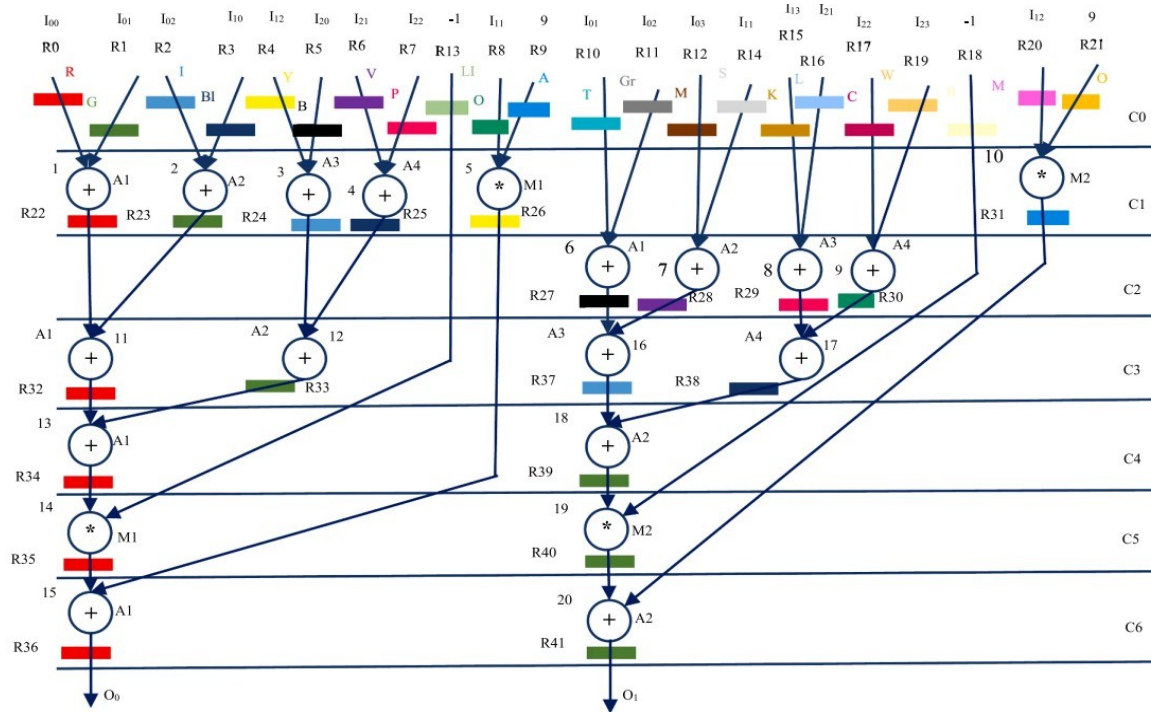


Figure 15: PSO-based architecture exploration of 4(+), and 2(*) resources used for scheduling 3*3 sharpening filter

Final register allocation table after embedding of security constraints

Table 18 Register allocation table (both before and after) in one table:

	C0	C1	C2	C3	C4	C5	C6
Red(R)	R0	R22	R22	R32	R34	R35	R36
Green (G)	R1	R23	R23	R33	R39	R40	R41
Indigo (I)	R2	R24	R24	R37	-	-	-
Blue (BL)	R3	R25	R25/R28	R38	R34	-	R36
Yellow (Y)	R4	R26	R26/R27	R26	R26	R26	-
Black (B)	R5	R26	R27/R26	R26	R26	R26	-
Violet (V)	R6	R25	R28/R25	-	-	-	-
Pink (P)	R7	-	R29	-	-	-	-
Lime (LI)	R13	R13	R13	R13	R13	-	-
Olive (O)	R8	R8	R30	-	-	-	-
Aqua (A)	R9	R31	R31	R31	R31	R31	-
Teal (T)	R10	R10	-	-	-	-	-
Gray (G)	R11	R11	-	-	-	-	-
Maroon (M)	R12	R12	-	-	-	-	-
Silver (S)	R14	R14	-	-	-	-	-
Khaki (K)	R15	R15	-	-	-	-	-
Lavender (L)	R16	R16	-	-	-	-	-
Crimson (C)	R17	R17	-	-	-	-	-
Wheat (W)	R19	R19	-	-	-	-	-
Beige (B)	R18	R18	R18	R18	R18	-	-
Magenta (M)	R20	R22	R22	R32	-	-	-
Orange (O)	R21	R24	R24	-	-	-	-

Results

Table 19: Area, latency, cost, and resource configuration of proposed hardware security methodology before and after embedding signature

Benchmarks	Explored resource configuration	Baseline unsecured design (before signature embedding)			Signature embedded secured design			Design cost overhead %
		Design area (um ²)	Design latency (ps)	Design cost	Design area (um ²)	Design latency (ps)	Design cost	
Blur filter (BF)	3(+), 1(*)	147.849	927.399	-0.112	147.849	927.399	-0.112	0
Sharpening filter (SF)	4(+), 2(*)	243.79	794.91	-0.193	243.79	794.91	-0.193	0
Laplace edge detection filter (LED)	2(+), 2(*)	199.75	728.671	-0.108	199.75	728.671	-0.108	0
Vertical embossment (VE)	1(+), 1(*)	99.09	596.185	-0.077	99.09	596.185	-0.077	0
Horizontal embossment (HE)	1(+), 1(*)	99.09	596.185	-0.077	99.09	596.185	-0.077	0

Results:

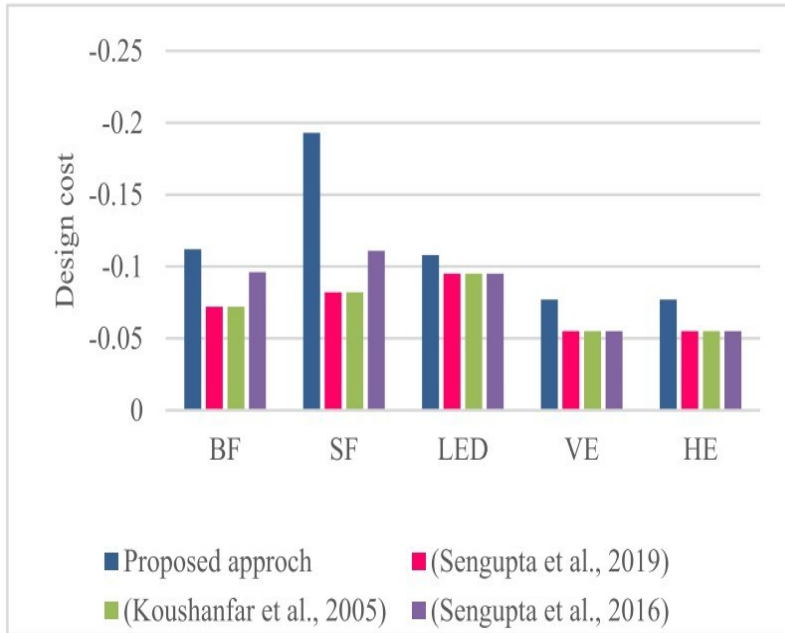


Figure 16: Design cost comparison between the proposed approach and (Sengupta et al., 2019) [5], (Koushanfar et al., 2005) [3], and (Sengupta et al., 2016) [4]

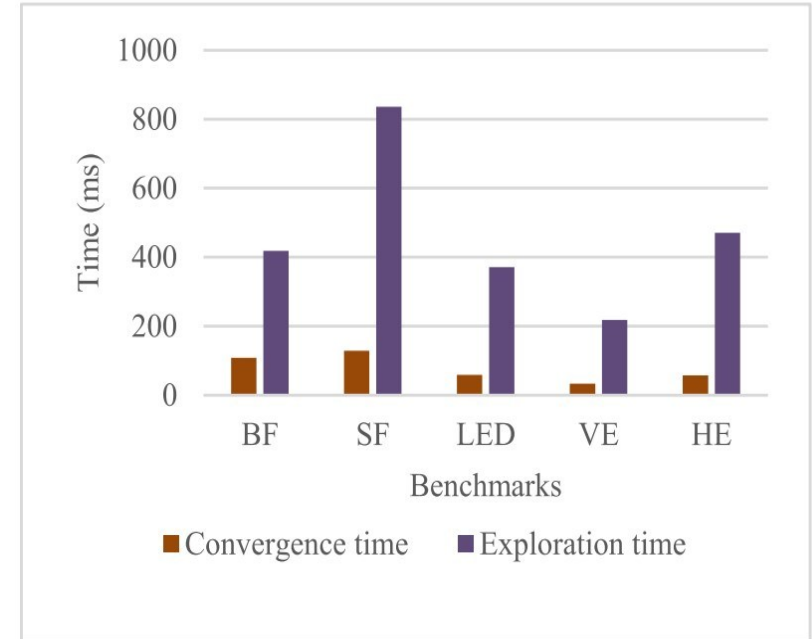


Figure 17: Convergence time and exploration time for the proposed methodology

Results (Pc comparison)

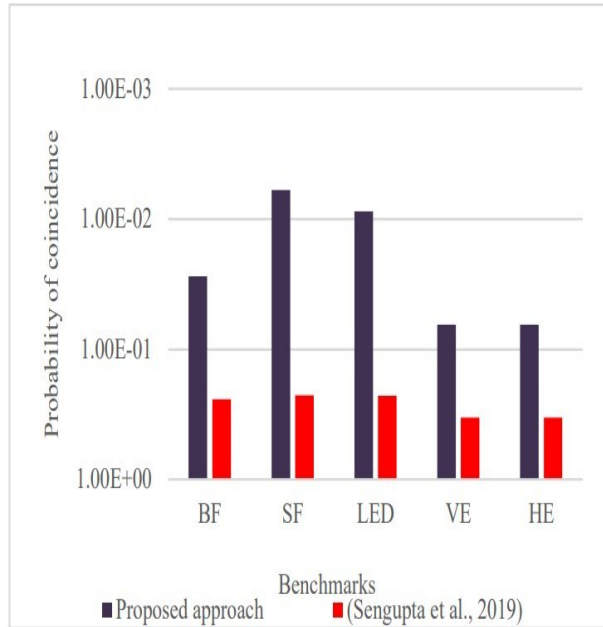


Figure 18: Comparison of probability of coincidence between the proposed and (Sengupta et al., 2019) [5]

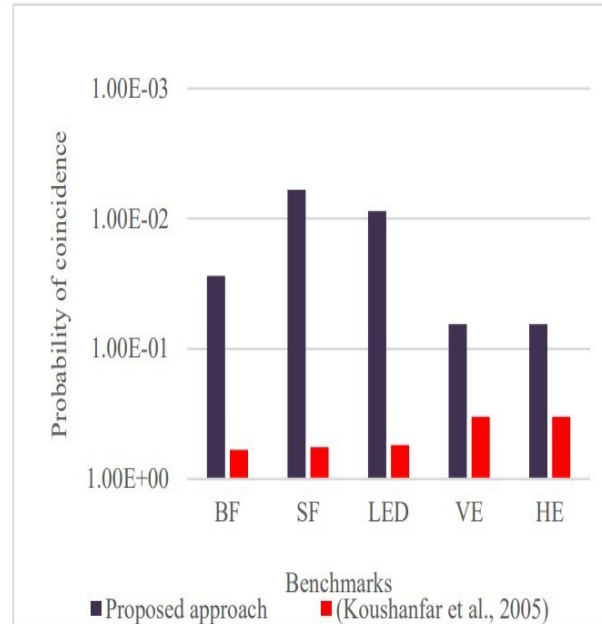


Figure 19: Comparison of probability of coincidence between the proposed and (Koushanfar et al., 2005) [3]

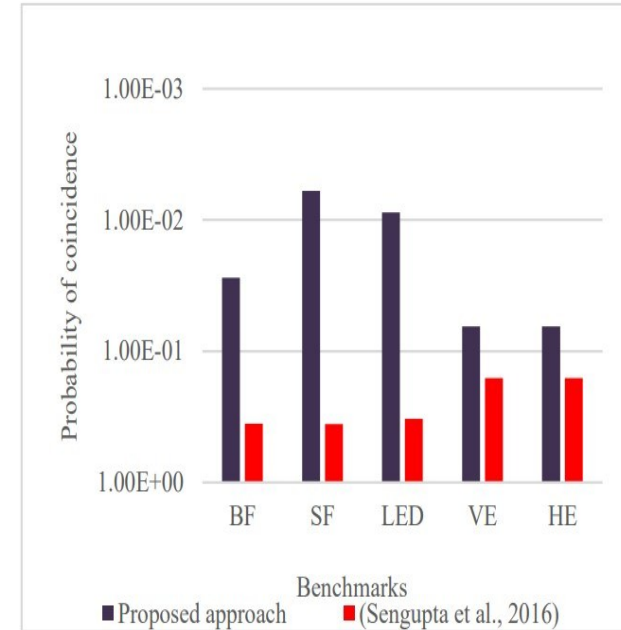


Figure 20: Comparison of probability of coincidence between the proposed and (Sengupta et al., 2016) [4]

Results (TT comparision)

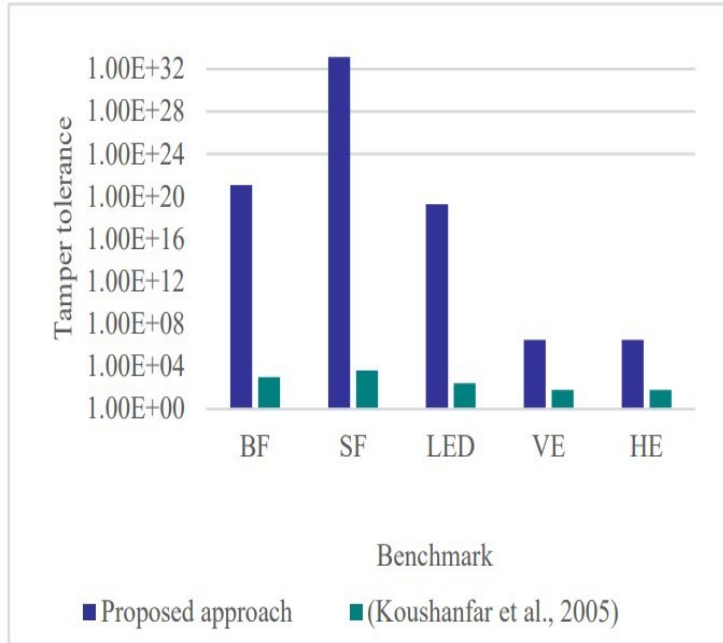


Figure 21: Comparison of tamper tolerance between the proposed and (Koushanfar et al., 2005) [3]

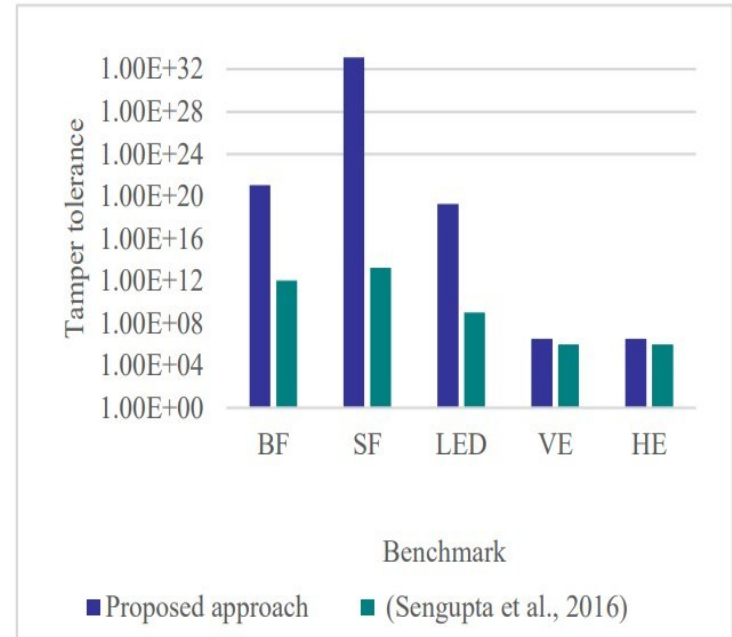


Figure 22: Comparison of tamper tolerance between the proposed and (Sengupta et al., 2016) [4]

Publication list (Journals):

Journal publications (12 publications):

1. A. Sengupta, R. Chaurasia and A. Anshul, "Robust Security of Hardware Accelerators Using Protein Molecular Biometric Signature and Facial Biometric Encryption Key," **IEEE Transactions on Very Large Scale Integration (VLSI) Systems**, vol. 31, no. 6, pp. 826-839, June 2023, *Impact Factor: 2.8*.
2. M. Rathor, A. Sengupta, R. Chaurasia and A. Anshul, "Exploring Handwritten Signature Image Features for Hardware Security," **IEEE Transactions on Dependable and Secure Computing**, vol. 20, no. 5, pp. 3687-3698, 1 Sept.-Oct. 2023, *Impact Factor: 7.0*.
3. A. Sengupta, A. Anshul, Secure hardware IP of GLRT cascade using color interval graph based embedded fingerprint for ECG detector. **Nature Scientific Reports**, 14, 13250 (2024), *Impact factor: 3.8*.
4. M. Rathor, A. Anshul, Anirban Sengupta, "Securing Reusable IP Cores using Voice Biometric based Watermark", **IEEE Transactions on Dependable and Secure Computing**, Accepted, Sep 2023, *Impact factor: 7.0*.
5. A. Sengupta, A. Anshul, V. Chourasia and N. Kumar, "M-HLS: Malevolent High-Level Synthesis for Watermarked Hardware IPs," **IEEE Embedded Systems Letters**, Accepted, June 2024, *Impact factor: 1.7*.
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Publication list (Journals):



Journal publications (Contd.):

9. A. Anshul, A. Sengupta, PSO based exploration of multi-phase encryption based secured image processing filter hardware IP core datapath during high level synthesis, **Elsevier Expert Systems with Applications**, Volume 223, 2023, 119927, ISSN 0957-4174, *Impact Factor: 7.5*.
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12. M. Rathor, A. Anshul, K. Bharath, R. Chaurasia, A. Sengupta, Quadruple phase watermarking during high level synthesis for securing reusable hardware intellectual property cores, **Elsevier Computers and Electrical Engineering**, Volume 105, 2023, 108476, ISSN 0045-7906, *Impact factor: 4.0*.

Publication list (Conferences):

Conference publications (7 publications):

1. A. Sengupta, R. Chaurasia and A. Anshul, "Hardware Security of Digital Image Filter IP Cores against Piracy using IP Seller's Fingerprint Encrypted Amino Acid Biometric Sample," **Proceedings of 8th IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST)**, Tianjin, China, 2023, pp. 1-6.
2. A. Sengupta, A. Anshul, S. Thakur and C. Kothari, "Fusing IP vendor Palmprint Biometric with Encoded Hash for Hardware IP Core Protection of Image Processing Filters," **Proceedings of 35th IEEE International Conference on Microelectronics (ICM)**, Abu Dhabi, United Arab Emirates, 2023, pp. 218-221.
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5. A. Sengupta and A. Anshul, "Key-Driven Multi-Layered Structural Obfuscation of IP cores using Reconfigurable Obfuscator based Network Challenge and Switch Control Logic," **Proceedings of 9th IEEE International Symposium on Smart Electronic Systems (iSES)**, Ahmedabad, India, 2023, pp. 141-146.
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Publication list (Book Chapters):

Book Chapters:

1. Anirban Sengupta, Aditya Anshul "Palmpoint Biometrics Vs. Fingerprint Biometrics Vs. Digital Signature using Encrypted Hash: Qualitative and Quantitative Comparison for Security of DSP coprocessors", *IET Book "Physical Biometrics for Hardware Security of DSP and Machine Learning Coprocessors"*, 2023.
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5. Aditya Anshul, Anirban Sengupta "Role of Consumer Technology and Connected Electronic Devices on SCM: A Discussion on its Usages, Impact, and Challenges", *UTHM Book "Evolution of Information, Communication and Computing System"*, 2023.

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1. A. Anshul and A. Sengupta, "A Survey of High Level Synthesis Based Hardware Security Approaches for Reusable IP Cores [Feature]," *IEEE Circuits and Systems Magazine*, vol. 23, no. 4, pp. 44-62, Fourthquarter 2023.
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3. F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection," *ACM Trans. Design Autom. Electron. Syst.*, vol. 10, no. 3, pp. 523–545, Jul. 2005.
4. A. Sengupta and S. Bhadauria, "Exploring Low Cost Optimal Watermark for Reusable IP Cores During High Level Synthesis," *IEEE Access*, vol. 4, pp. 2198-2215, 2016.
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15. A. Sengupta and R. Chaurasia, "Securing IP Cores for DSP Applications Using Structural Obfuscation and Chromosomal DNA Impression," in *IEEE Access*, vol. 10, pp. 50903-50913, 2022.



Thank You!